DARREN BOGGS

Engineering Technology Student

Contact

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About

I am a MET student with two years of engineering internship experience supporting equipment design, process improvements, and safety initiatives in a production environment. I enjoy taking ownership of technical projects, working directly with operators, and developing practical solutions that make a measurable impact.

Skills

· SolidWorks
· Autodesk Inventor
· AutoCAD
· CATIA
· Tooling Design
· PLC Programming
· Studio 5000
· FactoryTalk View
· Prototype Development
· Finite Element Analysis (FEA)
· P&ID Interpretation
· Process Improvement
· Lean Manufacturing
· Six Sigma Fundamentals
· Time Studies
· Data Collection & Analysis
· Safety Risk Assessment
· Technical Documentation
· Cross-Functional Collaboration
· Operator Engagement
· Mechatronics

Engineering Portfolio

Education

Tennessee Technological University | Bachelor of Science (BS)

Manufacturing Engineering Technology (MET) | Graduation: Dec '26
Concentration in Mechatronics | Minor in Business

Work Experience

Project Engineering Intern

May '25 – April '26

Bonnell Aluminum · Carthage, TN, USA

- Designed and validated a 4,000-lb load-rated production rack using CAD and FEA, meeting strict stability and safety requirements while supporting paint department operations.
- Developed a high-capacity exhaust filtration concept utilizing a 15 HP, 16,000 CFM fan system to reduce hazardous particulate discharge while balancing energy efficiency and maintenance demands.
- Led cross-functional equipment and safety improvement initiatives by collaborating with operators, maintenance, and engineering to implement practical, floor-ready solutions.
- Conducted time studies and process evaluations to analyze workflow and recommend improvements.
- Created detailed drawings, documentation, and presentations to support project implementation.

Controls Engineering Intern

May '26 – Aug '26

Snap-on · Elizabethton, TN, USA

- Designed and implemented a controls upgrade replacing a mechanical cam system with five sensors and a Turck TBEN remote I/O block.
- Programmed and modified Allen-Bradley PLC logic in Studio 5000 using timers, counters, one-shots, and bit manipulation instructions.
- Developed and updated HMI screens in FactoryTalk View Studio to provide operators with equipment status and sensor feedback.
- Developed CAD models and mechanical designs for frames, fixtures, equipment, and production improvement projects.
- Created detailed fabrication drawings, assembly drawings, and design documentation to support manufacturing and installation.
- Supported troubleshooting, installation, and testing of automated equipment through FAT/SAT and production-floor validation.

Certifications

Precision Measurement and Calibration Certification | Starrett

- Certified in the use of precision measuring and calibration instruments, including Starrett tape measures, rulers, and dial calipers.

OSHA 10-Hour | OHSA

- Completed the OSHA 10 hour general certification course.

FANUC Robotics CERT – In Progress | FANUC

- Training in FANUC industrial robot operation and programming, including safe setup, motion control, and automation fundamentals.